

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

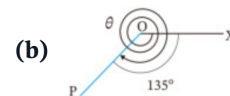
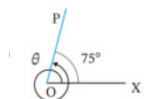
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Classified Practice

Defining Angles and Basic Trigonometric Functions

Q002 **Written****Draw the terminal side for 300° and -450° .**

Q003 Written

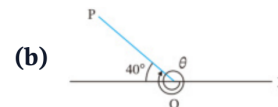
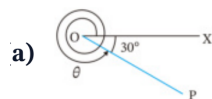
Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

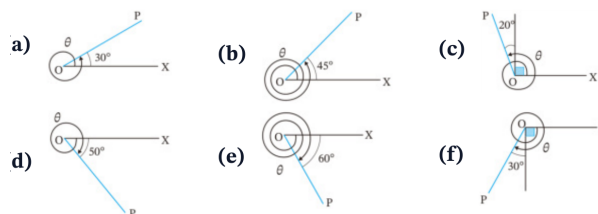
Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the source figure (a)–(f).

2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Q009 Written

Express $610^\circ, 920^\circ, 840^\circ, -430^\circ, -960^\circ, -20^\circ$ **as** $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Q011 MCQ

The principal angle coterminal with 765° is:**Choose the correct option.**A 45° B 135° C 225° D 315°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Choose the correct option.**A -50° B 50° C 310° D 410°

Q013 MCQ**The terminal side of 540° lies on the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Choose the correct option.**

A QI

B QII

C QIII

D QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Choose the correct option.**

- A $75^\circ + 360^\circ n$
- B $285^\circ + 360^\circ n$
- C $-285^\circ + 360^\circ n$
- D $360^\circ n - 75^\circ$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Choose the correct option.**

- A counter-clockwise
- B clockwise
- C radial only
- D undefined

Q017 MCQ**The terminal side of 1080° is the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Choose the correct option.**A $360^\circ n$ B $0^\circ n$ C $180^\circ + 360^\circ n$ D $90^\circ + 360^\circ n$

Q019 MCQ**The principal angle for -990° is:****Choose the correct option.**A 90° B 180° C 270° D -270°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Choose the correct option.

A 25°

B 155°

C 205°

D 335°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert 150° to radians:**Choose the correct option.**

- A $\frac{5\pi}{6}$
B $\frac{3\pi}{5}$
C $\frac{5\pi}{3}$
D $\frac{3\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Choose the correct option.**A -120° B -150° C 150° D -300°

Q030 MCQ

For $r = 5$, $\theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

- A $\frac{5\pi}{3}$
- B $\frac{10\pi}{3}$
- C $\frac{25\pi}{3}$
- D 10π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

- A π
- B 2π
- C 4π
- D 8π

Q032 MCQ

The radian measure of a full turn is:**Choose the correct option.**

- A π
 - B 2π
 - C 180
 - D 360
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

A 210°

B 270°

C 315°

D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

- A $\frac{\pi}{6}$
- B $-\frac{\pi}{6}$
- C $\frac{5\pi}{6}$
- D $\frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:**Choose the correct option.**

- A $\frac{3\pi}{2}$
B 3π
C 9π
D $\frac{9\pi}{2}$

Q036 MCQ**The angle $\frac{5\pi}{3}$ lies in:****Choose the correct option.**

A QI

B QII

C QIII

D QIV

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .

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Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q041 **Written**

Find the exact values of the eighteen trigonometric functions listed in the source exercise (a)–(r), including negative and undefined cases.

Q042 Written

The panel height is $h(t) = 3 \sin\left(\frac{2\pi}{3}t - \frac{\pi}{4}\right) + 1.5$. Find the exact height at $t = 0$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q043 MCQ

 $\sin \frac{\pi}{6}$ equals:

Choose the correct option.

A 0

B $\frac{1}{2}$ C $\frac{\sqrt{3}}{2}$

D 1

Q044 MCQ $\cos \frac{2\pi}{3}$ equals:**Choose the correct option.****A** -1**B** $-\frac{1}{2}$ **C** $\frac{1}{2}$ **D** $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:

Choose the correct option.

- A -1
- B 0
- C 1
- D undefined

Q046 MCQ $\sin\left(-\frac{\pi}{2}\right)$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:

Choose the correct option.

A 0

B 1

C -1

D undefined

Q048 MCQ $\cos \frac{5\pi}{3}$ equals:**Choose the correct option.**

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:

Choose the correct option.

A x

B y

C x/y

D $1/x$

Q050 MCQ $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Choose the correct option.**

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q052 Written

Find $\sin \frac{5\pi}{6}$, $\cos \frac{4\pi}{3}$, $\tan \frac{7\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q053 Written

Find $\sin(-\frac{5\pi}{6})$, $\cos(-\frac{7\pi}{6})$, $\tan(-\frac{\pi}{3})$.

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Q056 Written**State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the source Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 **Written**

Solve the sixteen equations in the source Exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Choose the correct option.

- A $\frac{\pi}{6}, \frac{5\pi}{6}$
B $\frac{\pi}{3}, \frac{2\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{5\pi}{3}$
B $\frac{2\pi}{3}, \frac{4\pi}{3}$
C $\frac{\pi}{6}, \frac{5\pi}{6}$
D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Choose the correct option.**

- A $\frac{\pi}{4}, \frac{5\pi}{4}$
- B $\frac{3\pi}{4}, \frac{7\pi}{4}$
- C $\frac{\pi}{4}, \frac{7\pi}{4}$
- D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $0, 2\pi$ D π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$ C π D $\frac{3\pi}{2}$

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{4\pi}{3}$
B $\frac{2\pi}{3}, \frac{5\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Choose the correct option.**

- A $\frac{\pi}{2}$
- B $\frac{3\pi}{2}$
- C $\frac{2}{\pi}$
- D 0

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:

Choose the correct option.

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C π
- D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:**Choose the correct option.**

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C $\frac{\pi}{4}, \frac{5\pi}{4}$
- D $\pi, 2\pi$

Q070 Written**Solve** $\cos \theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Q072 Written**Solve** $\tan \theta = \frac{1}{\sqrt{3}}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Q074 Written**Solve** $2 \sin \theta + \sqrt{2} = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

The radian measure of 240° is:

Choose the correct option.

- A $\frac{2\pi}{3}$
- B $\frac{4\pi}{3}$
- C $\frac{3\pi}{4}$
- D $\frac{5\pi}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{2\pi}{3}$
B $\frac{4\pi}{3}, \frac{5\pi}{3}$
C $\frac{3\pi}{2}, \frac{4\pi}{3}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The general-angle family for -300° is:

Choose the correct option.

- A $60^\circ + 360^\circ n$
- B $300^\circ + 360^\circ n$
- C $-60^\circ + 360^\circ n$
- D $120^\circ + 360^\circ n$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; 0 , π ; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; $\frac{5\pi}{4}$, $\frac{3\pi}{4}$; $\frac{7\pi}{4}$, $\frac{5\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{11\pi}{6}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{3\pi}{2}$, $\frac{15\pi}{6}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{2}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$